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# EECE2310

## Introduction to Digital Design and Computer Architecture

Course Syllabus – Summer II 2025

(Syllabus and Schedule are subject to change)

## Course Description

**Overview**:

Covers the fundamentals of digital design topics including digital data representations and operations as well as combinational and sequential circuits design. Introduces various topics of computer organization, architecture, and microarchitecture applicable to modern computer engineering systems. Discusses principles of computer memory addressing. Applies the concept of hardware and software interface through programming in assembly language.

**Course units**: 4 Credit Hours

**Prerequisite**: CS 1800 **Corequisite**: EECE 2311

**Course Learning Outcomes**:

1. Use combination and sequential logic to build circuits that solve complex computer engineering problems.
2. Assess the technical requirements implicit in a natural description of a problem and satisfy them with a hierarchical interconnection of combinational and sequential logic blocks.
3. Solve computer engineering problems susceptible of being solved with finite state machine models, and systematically translate them to sequential synchronous circuits.
4. Identify a computer processor datapath and control with respect to the instruction set and hardware organization.
5. Explain the concept of abstraction and standardization through Instruction Set Architecture definition.

**Recommended Textbooks:**

- S. Harris and D. Harris, "Digital Design and Computer Architecture: RISC-V Edition", 1st Edition, Morgan Kaufmann Publisher, 2021.
- M. M. Mano and M. D. Ciletti, "Digital Design", Pearson Publisher, 6th Edition, 2017.

**Course grades:**

The course grade will be determined according to the following weighs:

| | |
|---|---|
| Homework Assignments | 35% |
| Class Participation | 5% |
| Highest two grades of two Midterm and Final Exams | 60% |

The course grade letter will be determined according to the following distribution:

| A | A- | B+ | B | B- | C+ | C | C- | D+ | D | D- | F |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 100 93 | 90 | 85 | 80 | 75 | 70 | 65 | 60 | 57 | 53 | 50 | 0 |

## Course Instructor


**Prof. Emad Aboelela**

e.aboelela@northeastern.edu

617-373- 3386

Dana Research Center - Room 404


Office Hours:

In person and over Zoom:

MTWR: 12:00 – 01:00 pm

or by appointment

(no office hours on vacation days or after the last day of classes)

## Lecture Meetings

MTWR: 1:30 – 3:10 pm

Shillman Hall – # 325

## Lab Meetings

TR: 9:30 – 11:30 am

Hayden Hall – # 009D

## Course Assistant


**Vaishnavi Patil**

patil.vaish@northeastern.edu


Zoom Office Hours (Link on Canvas):

MTWR: 6:00 – 7:30 PM

## Lab Assistants


**Xiuhan Chen**

chen.xiuh@northeastern.edu

**Manas Modi**

modi.manas@northeastern.edu


Open Lab Hours: posted on Canvas

# Course Methodologies

**Class Meetings**: The course is taught exclusively in person and no remote participation is allowed. Attendance is mandatory and will be taken at every class meeting. This course may be recorded for educational purposes. Eating and drinking are prohibited in the classroom. As the instructor is lecturing, electronic devices, such as personal cell phones and laptops, are not allowed to be used unless specifically required to accomplish a course-related activity with the permission of the instructor. If you arrive at any class session late by more than **10 minutes**, you will be considered absent from that session. Violating the class meeting rules will result in points deducted from your class participation grade.

**Homework Assignments** will help students practice the theory presented in class. All assembly programs must be written in RISC-V. Submitted assembly programs must assemble and run correctly in the RARS Assembler. Submission of an assignment includes a PDF report along with any required RISC-V assembly programs following the assignments guidelines. Only submissions on Canvas are accepted.

**Submissions Policy:** Late submission of any assignment's deliverables after the given due date and time; will be penalized at 1 point per every late hour. No submission will be accepted 48 hours after its due date/time or after the assignment solutions are provided by the instructor, whichever comes first. There is no exception to this late submission policy. Therefore, make sure to check that all the required deliverables for any assignment are submitted correctly on Canvas before the deadline.

**Exams:** There will be two midterm exams during the weeks specified in the schedule at the end of this syllabus. Each midterm exam will have a duration of 90 minutes. The Final exam is a 2-hour exam and will be scheduled by the registrar office. The final exam is comprehensive. One of these three exams is optional as the best two scores from the two midterms and the final exams will be considered towards your course grade. No make-up exams will be allowed.

**TRACE**: Your participation in the anonymous and confidential course TRACE survey is very important as I will use your feedback to improve the future course implementation. Therefore, completing the survey will be considered part of your class participation activities. Proof of TRACE completion can be a copy of either the confirmation e-mail you will receive, or the info posted on your account under "My Evaluations" from the drop-down menu under the Home tab, on the right-hand corner.

**Plagiarism and Cheating**: Deliberate plagiarism in **any** assignment or cheating in exams are extremely serious offenses that may result in failing the course. Plagiarism includes submitting answers or programming codes that are not yours (e.g., copied from or provided to you by any source). Copying code from the slides of this course, which are posted on Canvas, is not considered plagiarism**.**

**Class Participation** is an essential part of the professional education process. In addition to following the class meeting rules and submitting the TRACE survey, a major part of the participation grade is assigned to answering the Poll Everywhere questions. Poll Everywhere submissions will be allowed only during the class meetings times, and it is a violation of the class policy to access the course Poll Everywhere activities while you are not attending the class in person. It is a violation of the Academic Integrity Policy to use any online source to search for answers to the Poll Everywhere questions. Unless otherwise is announced by the instructor at the time the questions are asked, you will not be evaluated on the correctness of your submitted answers on Poll Everywhere rather, on submitting relevant answers to the questions. Students who receive a **written pre-approved** excuse from attending a class due to an emergency are still required to watch the video of the missing lecture and email the instructor with the answers to its Poll Everywhere questions within **48 hours** of the lecture time. Such excuses are limited to **three times** per semester unless an official excuse letter from the university is submitted in advance.

**Course Assistants**: Their responsibilities include holding weekly office hours and grading the assignments and midterm exams.

# Course Policies and Resources

**Academic integrity:** A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University. As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in the Code of Student Conduct (available here) or by the direction of instructors. Any violation of academic integrity will be punished according to the university's policies and standards (available here).

**Title IX:** Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender identity, in educational programs and activities that receive federal financial assistance. Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty, and staff. In case of an emergency, please call 911. For a complete list of reporting options and resources both on- and off-campus, please click here.

**Accommodations**: Northeastern University and the Disability Access Services (DAS) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive service through the DAS, students must provide appropriate documentation that demonstrates a current substantially limiting disability. Mental health support is available at any time from any location through Find@Northeastern for all full-time degree-seeking students. Learn more from the Health and Counseling Services.

**Online Resources**: This course utilizes the university Canvas Course Management System. Through Canvas, you can access the course lectures, Panopto videos, homework assignments, Discussions, and Zoom rooms. The Panopto Videos platform will host the videos of the course lectures that might be recorded. Canvas Discussions tool facilitates interaction among students and instructors in an efficient and intuitive manner. Discussions is highly catered to getting help fast and efficiently from classmates, the instructor, and the course assistants. Rather than emailing questions to the teaching staff, it is encouraged that you post your questions on Discussions. Emails must be sent from your NU email address and must include your ID and course/section info. Expect replying to your online communications to take up to two business days. You must check, at least once a day, the course Canvas site, Announcements, Discussions, and your email for any updated contents, deadlines, or announcements. Also, make sure that Canvas notifies you whenever there are course-related updates or posts. For Notification Settings, click on your Canvas account icon (near the top-left corner) then click on Notifications. Zoom will be utilized for the office hours of the instructor and the course assistants as scheduled above. Zoom will be limited only to Northeastern students registered in Zoom with their official Northeastern email address.

| Tentative Course Outline | | | |
|---|---|---|---|
| **Week*** | **Lecture** | **Exams** | **Assignments** |
| **1** (06/30/2025) | 1. Introduction to Computer Systems<br>2. Digital Logic Gates<br>3. Digital Data Representations and Operations | | - Homework 1 |
| **2** (07/07/2025) | 4. Digital Logic Minimization<br>5. Combinational Digital Circuits<br>6. Sequential Digital Circuits | | - Homework 2 |
| **3** (07/14/2025) | - Sequential Digital Circuits (cont'd)<br>7. Registers and Counters<br>8. Overview of FPGA Design<br>- 1st Review | | |
| **4** (07/21/2025) | 9. RISC-V Computer Architecture | 1st Midterm Exam | - Homework 3 |
| **5** (07/28/2025) | - RISC-V Computer Architecture (cont'd)<br>10. Single-Cycle RISC-V Computer Microarchitecture | | - Homework 4 |
| **6** (08/04/2025) | - Single-Cycle RISC-V Computer Microarchitecture (cont'd)<br>11. Assembly Programming in RISC-V<br>- 2nd Review | | |
| **7** (08/11/2025) | - Assembly Programming in RISC-V (cont'd)<br>- Final Review<br>- Friday (08/15/2025): Reading Day (no class) | 2nd Midterm Exam | |
| **8** (08/18/2025) | Final Exam: TBD | | |

* The dates in this column are the dates of the Monday of each week. They are not necessarily the dates of any lecture, assignment, or exam that appear on their same row.